APPLIED MACHINE LEARNING · COURSE PROJECT 4

Classification with Non-Linearity, Tree-Based Methods & SVM

Bank Marketing Dataset

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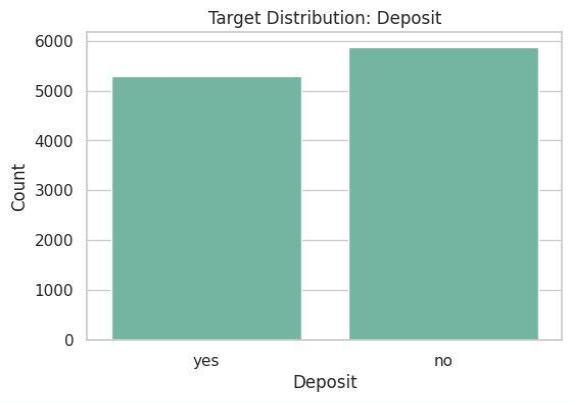
Problem Setup & Dataset



Near-balanced classes: ~47% yes / ~53% no

Feature Types

- **Profile: age, job, marital, education**
- **Financial: balance, default, housing, loan**
- **Campaign: contact, day, month, campaign, duration**
- **History: previous, pdays, poutcome**

Dataset Overview

n = 11,162

Binary target: deposit ∈ {0,1}

Predictors: numeric + categorical

Predict whether a client subscribes to a term deposit

Each row = one client-contact case from a bank telemarketing campaign

Feature Interpretations

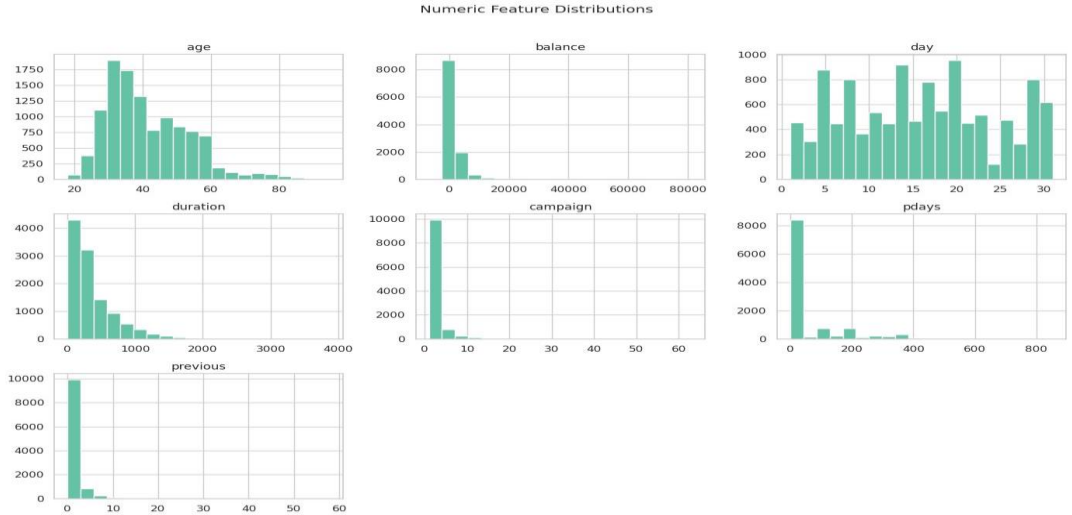
campaign: how many times the client was contacted in the current campaign
previous: how many contacts happened before this campaign
pdays: days since the last previous contact
Poutcome: outcome of the previous campaign
duration: length of the last call
deposit: final subscription outcome

Preprocessing Steps

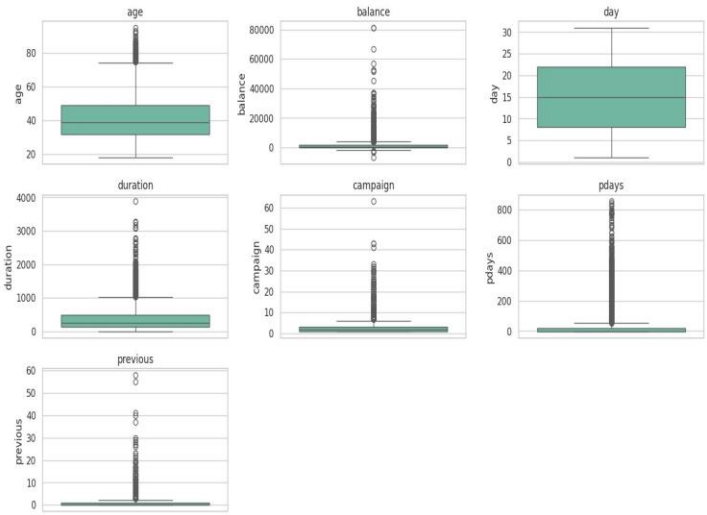
- Dropped *duration* (data leakage)
- One-hot encoding for categoricals
- Robust scaling for sensitive models
- *Engineered: was_previously_contacted = $\mathbb{1}(pdays \neq -1)$*

Exploratory Data Analysis

Numeric Feature Distributions



Outlier Profiles (Boxplots)



Key EDA Findings:

- balance & pdays are heavily right-skewed with extreme outliers
- campaign and previous are sparse count variables
- age is roughly bell-shaped, median ~38
- duration excluded to prevent leakage

Motivation: Why Non-Linearity?

Linear Logistic Regression

$$\log(p/(1-p)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Assumption: decision boundary is a hyperplane in feature space.

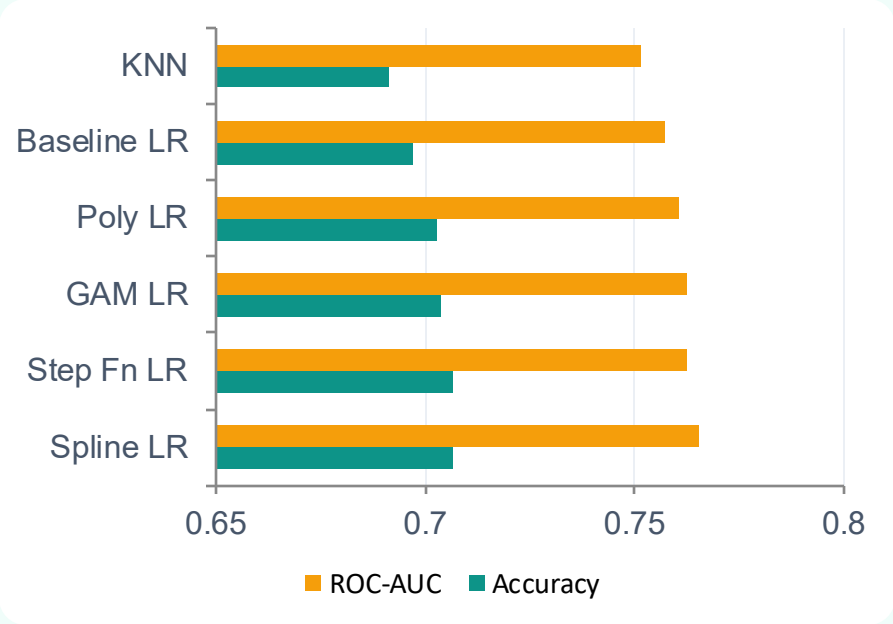
Non-Linear Extension

$$\log(p/(1-p)) = \beta_0 + \Sigma\, \beta_j\, \varphi_j(x)$$

Replace raw inputs with basis functions $\phi_j(x)$ to capture curvature.

Method	Basis Function $\phi(x)$	Key Property
Polynomial	$x, x^2, x^3, \ldots$	Global, smooth
Step Function	$\mathbb{1}(x \in B_k)$	Piecewise constant
Spline	Smooth piecewise polynomials	Local + smooth
GAM	$f_1(x_1) + f_2(x_2) + \cdots$	Additive structure
KNN	Majority vote of k neighbors	Non-parametric

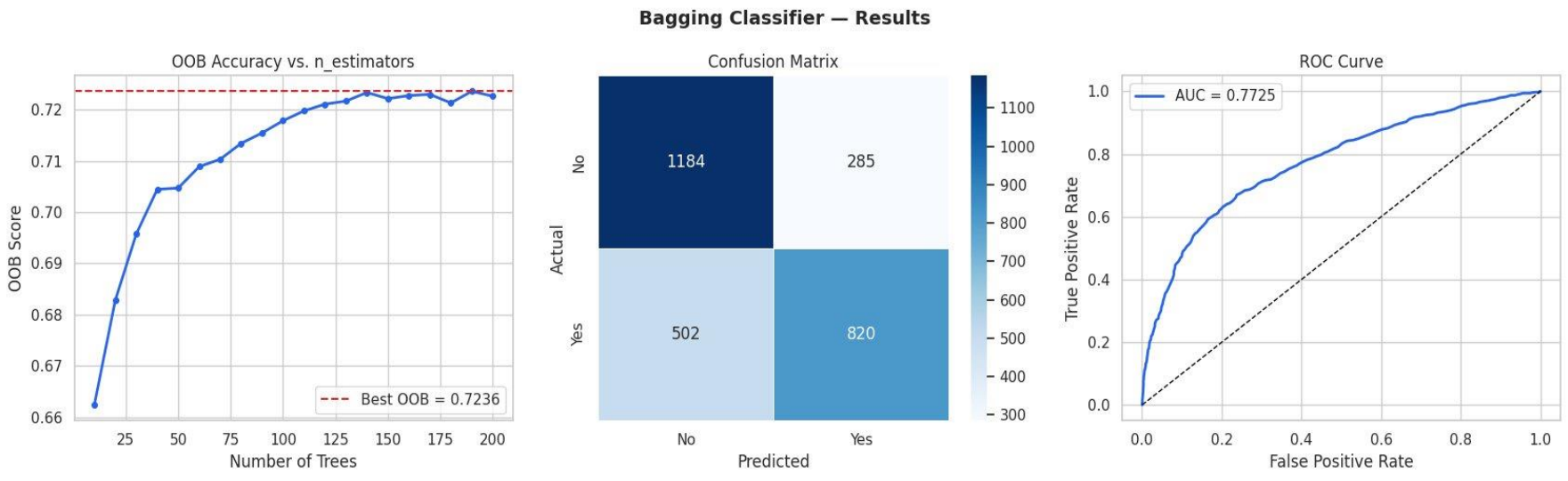
Part 1: Non-Linear Models — Results



Model	Acc	F1	AUC
Spline Logistic ★	0.7069	0.6510	0.7653
Step Function LR	0.7066	0.6543	0.7626
GAM-style LR	0.7040	0.6461	0.7627
Polynomial LR	0.7026	0.6438	0.7608
Baseline LR	0.7026	0.6369	0.7573
KNN	0.6912	0.6354	0.7516

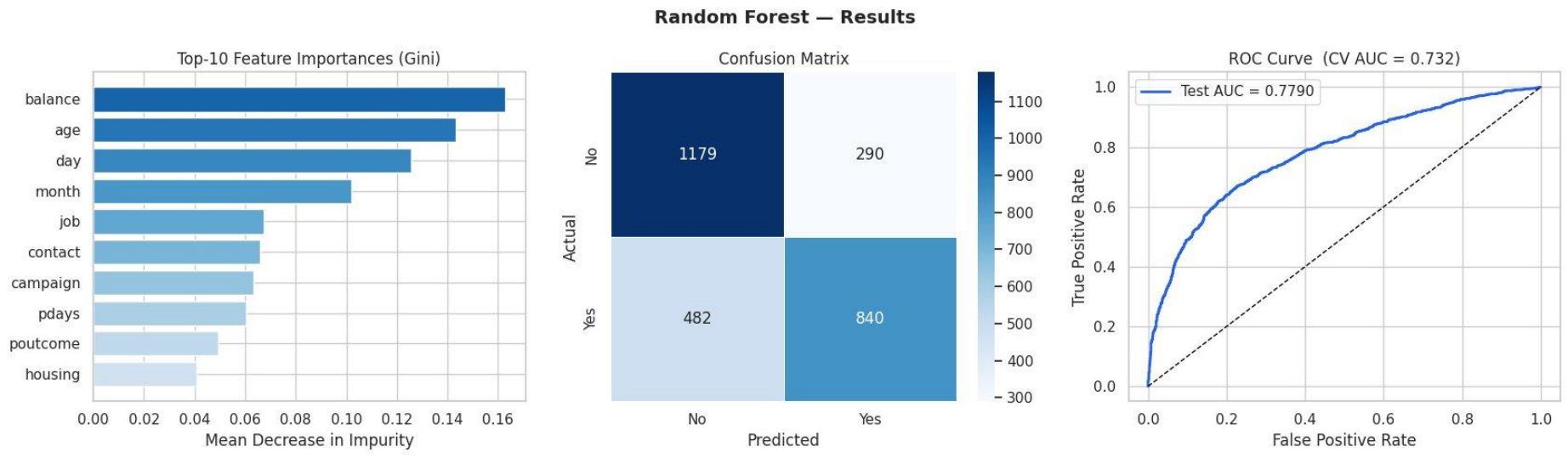
Key takeaway: Smooth non-linear transformations consistently beat the linear baseline. Gains are moderate, indicating mild non-linearity in the data.

Part-2: Decision Tree based methods(Bagging)



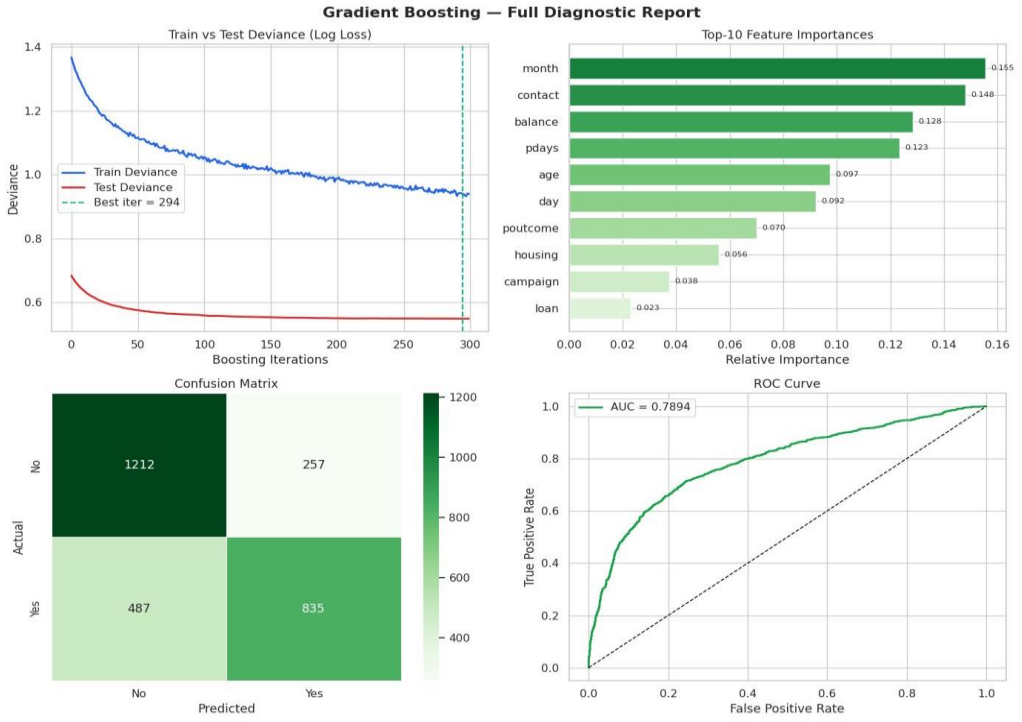
OOB Score converges at ~200 trees (best OOB 0.7236) · AUC = 0.7725 · Confusion matrix shows stronger recall on class No

Random Forest



balance, age, day, month top Gini importance · Test AUC = 0.7790 · CV AUC = 0.732 confirms generalization

Gradient Boosting



Best iteration = 294 trees · month & contact dominate · AUC = 0.7894 **Best overall model**

Tree-Based Ensemble Methods Result

Bagging

Bootstrap-aggregated trees.

Variance reduction via averaging:

$\hat{f}_{bag}(x) = (1/B) \Sigma \hat{f}^{*b}(x)$

Base: fully grown decision trees.

Random Forest

Bagging + random feature subset

$m \approx \sqrt{p}$ features at each split.

Decorrelates trees → lower variance.

Built-in OOB error estimate.

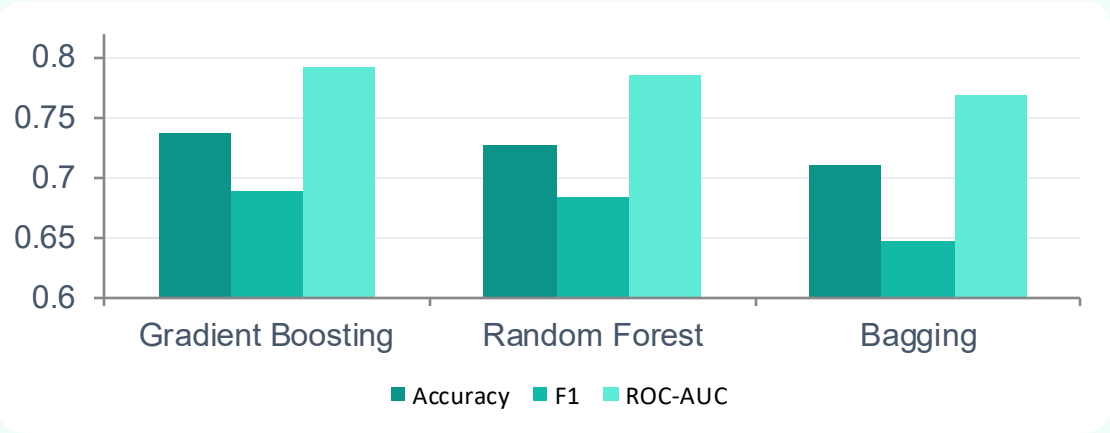
Gradient Boosting

Sequential additive models.

Each tree fits residuals:

$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$

Shrinkage ν controls learning rate.



Results

Gradient Boosting

Acc:0.7367 F1:0.6890 AUC:0.7911

Random Forest

Acc:0.7263 F1:0.6830 AUC:0.7846

Bagging

Acc:0.7098 F1:0.6475 AUC:0.7686

CONCLUSION

Best Final Model:

Gradient Boosting Classifier

Acc: 0.7367 · F1: 0.6890 · ROC-AUC: 0.7911

1. Non-linearity matters → smooth transforms outperform linear logistic regression
2. Kernel choice matters → RBF is best; Sigmoid collapses
3. Tree ensembles win → they capture both non-linearity and feature interactions
4. Campaign timing (month, contact) is the dominant predictive signal

Part 3: Support Vector Machines

SVM Primal & Kernel Trick

Maximize margin: $2 / \|w\|$

Subject to: $y_i (w^T x_i + b) \geq 1$

Kernel trick avoids explicit $\varphi(x)$:

$K(x_i, x_j) = \langle \varphi(x_i), \varphi(x_j) \rangle$

Soft-margin: penalizes slack ξ_i with hyperparameter C.

Linear

$K = x^T y$

Acc: 0.6643 AUC: 0.7328

Fastest; may underfit

Polynomial

$K = (\gamma x^T y + r)^d$

Acc: 0.7094 AUC: 0.7672

Captures degree-d interactions

RBF ★

$K = \exp(-\gamma \|x-y\|^2)$

Acc: 0.7152 AUC: 0.7687

Best; infinite-dim mapping

Sigmoid

$K = \tanh(\gamma x^T y + r)$

Acc: 0.5629 AUC: 0.5523

Weakest on this dataset

Final Comparison: All Models

Best: Gradient Boosting

Best SVM: RBF kernel

Best NL: Spline LR

Worst: SVM Sigmoid

Accuracy

Recall

Thank you!